

MUD LOGGING REPORT

REPORTING • INTERPRETATION • QUALITY CONTROL

Professional Wellsite Reporting Framework
for Drilling & Geological Operations

DAILY REPORT

GAS & GEOLOGY

COMPOSITE LOG

END OF WELL

PT. GEO OPTI OILFIELD

Reporting Objectives & Value

Provide a reliable daily record of drilling, geology, gas and mud-system conditions.

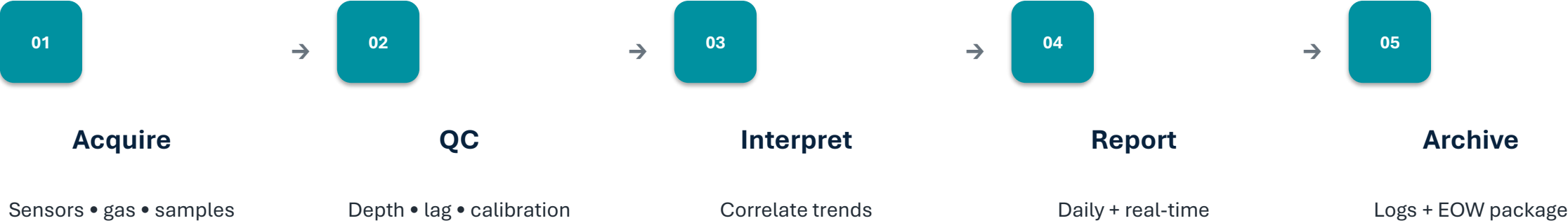
Convert real-time wellsite observations into concise operational information.

Highlight formation changes, hydrocarbon shows, pressure trends and significant events.

Maintain traceable data for drilling decisions, geological evaluation and final well documentation.

Standardize communication between Mud Logging, Drilling, Geology and remote support teams.

Mud Logging Reporting Cycle

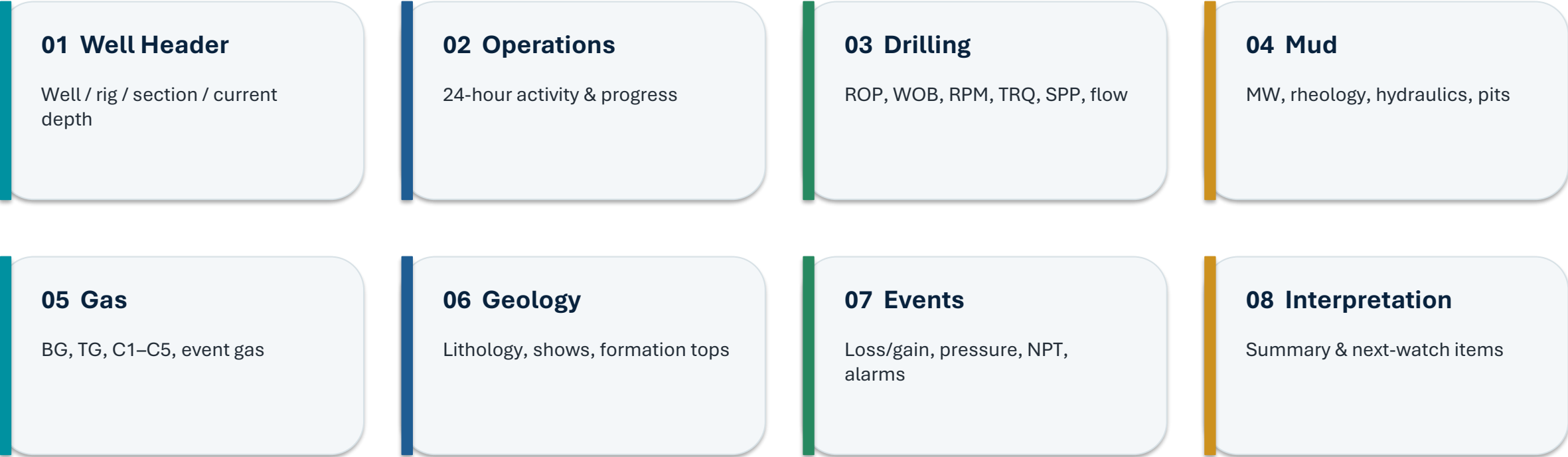


Daily Mud Logging Report – Structure

MUDLOGGING MORNING REPORT									
COMPANY WELL :					REPORT NUMBER : 20				
WELL NAME :					DATE : 18-Nov-25				
SPUD DATE : 2-Nov-25 18:00 WIT					TIME (05:00) AM :				
WELL CLASS :					MIDNIGHT DEPTH : 1928 MTR				
RIG NAME :					DEPTH : 1986 MTR				
RT - MSL ELEVATION : 56 MTR					TVD : 1986 MTR				
GL ELEVATION : 6 MTR					PREVIOUS DEPTH : 1822 MTR				
COMPANY MAN :					PROGRESS 15 HRS : 164 MTR				
GEOLOGIST :					CASING SHOE DEPTH 862 MTR				
ENGINEER :					LAST SURVEY :				
LOGGER :					DEVIATION Inc :				
UNIT NO :					Az :				
ACTIVITY (08:00-11) DOF 12.25" PDC									
					PREVIOUS BIT				
BIT NO : NB					BIT NO : NB				
JET SIZE / TFA : 1x12, 5x15, 1x16 1.170					JET SIZE / TFA : 2x14, 5x15 1.164				
TYPE : GT56S					TYPE : GT56S				
S/N : 14350884					S/N : 14267752				
IADC : -					IADC : -				
BIT 24 HRS : 12.90 HRS					BIT HRS : 32.3 HRS				
TOTAL BIT HRS : 67.80 HRS					DEPTH IN : 863 MTR				
DEPTH IN : MTR					DEPTH OUT : 1225 MTR				
DEPTH OUT : MTR					FOOTAGE : 362 MTR				
FOOTAGE : MTR					ROP : 4.8 Min/M				
ROP : Min/M					BIT COND : 1-3-BT-S-X-I-BU-PR				
BIT SIZE : 12.25"					BIT SIZE : 12.25"				
DRILLING DATA					PUMPS DATA				
ROP MAX : 1.8 Min/M					PUMP OUTPUT : 0.0852 BLS/STKS				
ROP MIN : 24 Min/M					SPM#1 : 51				
AVR ROP 24 HRS : 12.7 Min/M					SPM#2 : 62				
AVG. ROP TOT : 16.6 Min/M					SPM#3 : 73				
WOB : 10-15 KLBS					Tot. SPM : 186				
PUMP PRESSURE : 2850 - 3000 PSI					GPM : 665 GAL/MIN				
ROTARY RPM'S : 50 - 80					BIT HYDRAULICS				
MUD MOTOR RPM : 113 - 119					JET VELOCITY : 57 m/sec				
TORQUE : 1 - 14 klbft					BIT POWER / Tot HHP : 150.9 HP				
MUD PROPERTIES					JET IMP.FORCE : 804 LBS				
MUD TYPE : KCL Polymer					IMPACT / AREA : 6.8 sq.in				
MW IN / OUT : 1.47 / 1.47 SG					BIT / AREA : 1.28 hp/in2				
VISCOSITY : 66 CPS					PRESSURE LOSS @ BIT : 380 PSI				
PLASTIC VIS : 26 CPS					PERCENTAGE LOSS @ BIT : 13.64 %				
YIELD POINT : 30 LB/100FT ²					ANN VEL @ DP / DC : 41 m/min				
CL - : 65000					WELL GEOMETRY				
K+ : 48000					CASING SHOE DEPTH : 862 MTR				
WATER LOSS : 4.8 cc					OPEN HOLE FOOTAGE : 1124 MTR				
PH : 10.5									
MTI/MTO : 69/75 °C									
GAS DATA									
MAXIMUM GAS : 33.2 UNITS									
AVG BG : 20 UNITS									
CONNECTION GAS : - UNITS									
TRIP GAS : - UNITS									
WIPER TRIP GAS : UNITS									
SHORT TRIP GAS : UNITS									

MUDLOGGING MORNING REPORT													
DEPTH		ROP (Min/M)		CUTTING %									
MD	ROP RANGE	ROP AVG	COAL	CLST	SH	SLST	SST	LST	TUFF	BSMT	REMARKS		
GAS READINGS													
DEPTH		TG	C1		C2		C3		IC4	NC4	IC5	NC5	
MD (M RKB)		Units	ppm										
PEAKS GAS													
DEPTH		BG	TG	C1		C2		C3		IC4	NC4	IC5	NC5
MMD		Units		ppm									
ACTIVITY													
13-Nov-25													
From	To	Activity											
00:00	18:00	Lanjut Bor formasi Lubang 12-1/4" dari 1577 mMD spi 1677 mMD - Drill Par : WOB/GPM/SPP/RPM/RPMM/TRQ : 10-30 klbs/710-730/2920-3150 psi/50/115-118/1200-6700 ft.lbs (ROP Avg 8 Min/meter). - Last Lithology @1668 mMD: SH: 90%, SST: 10%. - BG/Max Gas : 5-10 U/ 34 U @1649 mMD : SH: 90%, SST: 10%. - Last survey MD: 1648.69 mMD/1648.03 mTVD, Incl: 9.58 deg, Azim: 283.60 deg, C-C: 8.91 m. - Intv slide : 1590 - 1604 mMD (14 meter/ 2.56 Jam). ROP Avg 11 Min/meter. - Intv slide : 1621 - 1630 mMD (9 meter/ 1.8 Jam). ROP Avg 12.5 Min/meter. - Intv slide : 1649 - 1663 mMD (14 meter/ 2.59 Jam). ROP Avg 11.1 Min/meter. - Bit Hours : 12.5 hrs, Kumm Bit Hours : 35.8 hrs - Dump possumbely per stands. - Pompa Low-vis 25 bbls, 1582 mMD, 1595 mMD, 1611 mMD, 1640 mMD. - Ditemukan kebocoran pada MP#2											
	18:00	Sirkulasi sambil perbaikan MP#2 dengan FR/SPP/RPM/M : 610/2230 psi/30/100 - Saat sirkulasi gas tinggi di kedalaman @1677 mMD: 911 U: SH: 90%, SST: 10%. - Perbaikan pada dumpener MP#2											
	20:00	Lanjut Bor formasi Lubang 12-1/4" dari 1677 mMD spi 1697 mMD - Drill Par : WOB/GPM/SPP/RPM/RPMM/TRQ : 10-20 klbs/700-720/3010-3025 psi/50/107-112/1300-6000 ft.lbs (ROP Avg 6.5 Min/meter). - Last Lithology @1694 mMD: SH: 90%, SST: 10%. - BG/Max Gas : 10 U/ 17 U @1695 mMD : SH: 90%, SST: 10%. - Last survey MD: 1677.56 mMD/1176.37 mTVD, Incl: 12.24 deg, Azim: 284.13 deg, C-C: 9.02 m. - Intv slide : 1679 - 1689 mMD (10 meter/ 2 Jam). ROP Avg 12 Min/meter. - Dump possumbely per stands. - Pompa Low-vis 25 bbls, 1582 mMD, 1595 mMD, 1611 mMD, 1640 mMD, 1697 mMD - Bit Hours : 3.05 hrs, Kumm Bit Hours : 38.85 hrs											

Daily Mud Logging Report – Structure



Well Header & Status

IDENTIFICATION

Well name / number
Rig & operator
Field / location
Report date / report no.
Prepared & checked by

CURRENT STATUS

Current MD / TVD
Hole size / section
Bit / BHA
Casing shoe
Current operation

PROGRESS

24-h drilled interval
Average / max ROP
Hours drilling
Hours non-drilling
Next planned activity

24-Hour Operations Summary

Summarize operations chronologically with depth and time references.

Separate drilling, connection, reaming, circulating, trip, casing/cementing and NPT.

Record major parameter changes and their operational reason.

Document significant geological or gas events at the corresponding lagged depth.

Keep the summary concise enough for management while retaining technical traceability.

Drilling Performance Reporting

DEPTH / ROP

Start–end depth
Average / max ROP

MECHANICS

WOB • RPM • Torque
Hookload / drag

HYDRAULICS

SPP • SPM • Flow
Pump configuration

EFFICIENCY

MSE trend
Connection time
NPT / dysfunction

Mud & Hydraulics Reporting

Report Mud Weight In/Out, funnel viscosity and key rheology where available.

Record flow rate, pump strokes, pit volume and trip-tank observations.

Annotate mud treatment, dilution, LCM, sweeps and pit transfers.

Correlate surface mud changes with gas extraction efficiency and pressure indicators.

Clearly distinguish measured values from calculated or estimated hydraulics.

Gas Data Reporting

ROUTINE GAS

Background Gas
Total / Maximum Gas
C1, C2, C3, iC4/nC4,
iC5/nC5

EVENT GAS

Connection Gas
Trip Gas
Swab Gas
Produced / recycled response

INTERPRETATION

Gas vs ROP
Gas vs lithology
Gas ratios
Normalized gas if used
Confidence / remarks

Gas Event Classification

Liberated / drilled gas — mechanically released while drilling the formation.

Produced gas — formation gas entering the drilling fluid when formation pressure exceeds effective wellbore pressure.

Recycled gas — previously observed gas returning after recirculation.

Contamination gas — gas related to additives, diesel/crude, mud products or active-pit operations.

Classification must use gas shape, timing, composition and operational context.

Geological & Cuttings Reporting

LITHOLOGY

Rock type & %
Color / texture
Grain size / sorting
Hardness / cement
Accessory minerals

SAMPLE QUALITY

Sample interval
Cavings
Contamination
Recovery quality
Lag confidence

INTERPRETATION

Formation tops
Reservoir indication
Seal / shale
Fracture / alteration
Pressure-related cavings

Hydrocarbon Show Reporting

Fluorescence: color, intensity, distribution and percentage.

Cut / solvent response and residual fluorescence where applicable.

Visible stain, odor or oil indication.

Gas response and C1–C5 composition associated with the show.

Lithology, porosity indication, interval thickness and show quality.

State whether the observation is direct, inferred or affected by contamination.

Lag Time & Sample Depth Control

Report calculated lag / bottoms-up in time and pump strokes.

Validate lag after changes in hole size, BHA, flow rate or major mud properties.

Assign cuttings and gas to the correct lagged depth.

Record tracer or operational validation when performed.

Flag uncertain lag so interpretation is not presented with false precision.

Pressure Surveillance Reporting

TREND INPUTS

d-Exponent
ROP / WOB
Gas trend
Cavings
Mud weight

SUPPORTING DATA

SPP / flow
Torque / drag
Drilling break
Shale observations
LWD/MWD if available

REPORTING

Trend change
Depth interval
Confidence level
Persons informed
Recommended surveillance

d-Exponent & Drilling Break

Plot d-Exponent / corrected d-Exponent consistently by bit run and formation.

Establish the local normal trend before interpreting a departure.

A drilling break should be correlated with lithology, gas and drilling parameters.

Avoid pressure conclusions from a single parameter; report converging evidence.

Annotate bit, BHA, drilling-mode and parameter changes that can alter the trend.

Well-Control / Significant Event Reporting

Record event time, drilled/lagged depth and current operation.

Capture pit gain/loss, flow behavior, gas response and relevant drilling parameters.

Document verification steps to exclude sensor error or planned pit transfer.

Record notification/escalation to Driller, Company Man and relevant disciplines.

Mud logging report documents surveillance and observations; well-control actions follow the rig's approved procedures.

Connection, Trip & Swab Gas

Connection Gas: report background, peak gas, ABG if used, expected lag and return to baseline.

Trip Gas: relate to trip interval, return to bottom and bottoms-up circulation.

Swab Gas: correlate with pipe movement and transient pressure reduction.

Recycled Gas: compare timing and shape with a prior response after circulation.

Include operational annotations so the gas event can be reconstructed later.

Loss / Gain Reporting

LOSS REPORT

Start depth/time
Loss rate / cumulative loss
Returns behavior
Formation/lithology
LCM / treatment
Recovery status

GAIN REPORT

Pit gain / flow increase
Pump status
Mud transfer verification
Gas / pressure indicators
Escalation
Final disposition

Composite / Master Log

DEPTH	ROP	LITHOLOGY	GAS	DRILLING	REMARKS
2500	20	SHALE	BG 7	WOB/RPM	Top
2525	35	SST	TG 25	SPP	Gas show
2550	58	SST/SH	TG 60	MSE	Pressure watch
2575	25 m/h	SHALE	BG 10	d-Exp	Stable

Real-Time Monitoring & Event Log

Maintain a time-based event log for connections, trips, pump-off, reaming, sweeps and alarms.

Use depth-based plots for formation, lithology, gas and drilling trends.

Synchronize real-time annotations with Daily Mud Logging Report entries.

Capture data gaps, sensor faults and calibration activities.

Use playback to verify event sequence before issuing final interpretation.

Daily Interpretation Summary

WHAT CHANGED?

Formation / lithology
ROP / drilling response
Gas / show
Pressure trend
Hole condition

WHY IT MATTERS

Reservoir potential
Pressure concern
Loss / stability risk
Drilling efficiency
Data-quality issue

WHAT TO WATCH NEXT

Expected formation
Gas baseline
d-Exponent trend
Pit / flow behavior
Sampling / QC priority

Recommended KPI Dashboard

KPI 1

24-h drilled footage / meters

KPI 2

Average & max ROP

KPI 3

Connection / slip-to-slip time

KPI 4

Sensor / data availability

KPI 5

Gas-system calibration status

KPI 6

Sample recovery coverage

KPI 7

NPT / significant events

KPI 8

MSE / d-Exponent trend status

Data Quality Control

Check sensor zero/span, calibration status and communication health.

Confirm gas trap condition, gas-line flow, degasser position and chromatograph response.

Validate lag time and depth synchronization.

Annotate pit transfers, mud treatment, diesel/crude additions, LCM and degasser changes.

Review spikes, flat-lines, drift and missing data before reporting.

Use checker/approver workflow for critical daily and final deliverables.

End of Well Report

EXECUTIVE SUMMARY

Well objectives, final depth, key events

GEOLOGY

Formation tops, lithology, reservoir/show summary

GAS

Gas zones, composition, ratios, event gas

DRILLING

Performance, MSE/ROP, NPT, dysfunction

PRESSURE / EVENTS

Pressure trends, losses/gains, significant events

APPENDICES

Final composite log, calibration, daily reports, data archive

Professional Reporting Standards

Use consistent units, abbreviations, depth references and time format.

Separate measured data, calculated values and interpretation.

Use neutral technical language; avoid unsupported certainty.

Every significant event should be traceable to time, depth and source data.

Charts must have readable scales, legends and operational annotations.

Apply document control: report number, revision, preparer, checker and approval.

Example Daily Executive Summary

DAILY EXECUTIVE SUMMARY

Drilled from 2,450 mMD to 2,610 mMD during the reporting period. ROP increased across a sandstone-dominant interval and was accompanied by a lag-corrected Total Gas increase above established background. C1–C5 response and fluorescence were recorded with the show interval. d-Exponent displayed a downward departure from the preceding trend; however, no interpretation is based on this indicator alone. Pit volume, flow-out and mud-weight behavior remained under continuous surveillance. The event, supporting plots and sample observations were communicated to the relevant wellsite personnel. Next-shift priorities are confirmation of formation continuity, gas baseline, lag accuracy and pressure-related trends.

Example wording only — values and interpretation must be replaced by actual well data.

Conclusion

A professional Mud Logging Report integrates drilling, gas, geology, mud and operational context. Daily reporting should highlight both the data and the interpretation behind significant changes. Depth/lag accuracy and gas-system QC are fundamental to reliable formation evaluation. Composite logs and event annotations provide the traceable technical record of the well. Consistent templates and QC improve communication from rig site to office and End-of-Well reporting.